What the algorithm does with each matrix

The algorithm is also a test. Feed it a matrix and how it terminates classifies the matrix — completion means positive definite, and each way of failing identifies a different reason.

5
CASES

THE ALGORITHM COMPLETES — 2

1 Symmetric positive definite

all $\lambda_i > 0$

$A = LL^{\mathsf{T}}$, all $\ell_{ii} > 0$

The intended case. Every square root is taken of a strictly positive number, the diagonal of L comes out positive, and the factorization is unique. Half the cost of LU because symmetry means only one triangle is computed.

2 Near-singular but definite

smallest λ near zero

runs; ℓ_{ii} very small

Completes without complaint while the accuracy quietly degrades — small pivots amplify rounding into L. The failure is silent, so the smallest diagonal entry is worth monitoring; adding εI regularizes at the cost of solving a slightly different problem.

THE ALGORITHM HALTS — AND THE HALT IS THE ANSWER — 3

3 Positive semi-definite

some $\lambda_i = 0$

zero appears on the diagonal of L

Rank-deficient rather than wrong. Pivoted Cholesky, $P^{\mathsf{T}}AP = LL^{\mathsf{T}}$, handles it by reordering so the zeros arrive last — the factorization exists but is no longer unique.

4 Symmetric indefinite

some $\lambda_i < 0$

negative under the square root

No real L exists, and the algorithm discovers this the moment it tries. Use LDL^{T}, which permits negative entries in D and so handles any symmetric matrix — at the price of no longer testing definiteness.

5 Not symmetric

fails before it starts

$A = LL^{\mathsf{T}}$ is impossible

Any product LL^{T} is symmetric by construction, so a non-symmetric A has no such factorization whatever its eigenvalues are. This is a precondition rather than an outcome — check symmetry first and use LU if it fails.

This is why Cholesky is the standard positive-definiteness test. Computing eigenvalues to check they are all positive costs $O(n^3)$ with an iterative method and gives approximate answers; running Cholesky costs $\frac{1}{3}n^3$, terminates in finitely many steps, and answers exactly — with the factorization in hand as a byproduct when the answer is yes.